DEVELOPMENT OF SMART PLANT WATERING SYSTEM

A PROJECT REPORT

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1. Title Page

Title: Smart Plant Watering System Using Soil Moisture Sensor and Automatic Water Pump Control

2. Abstract

Water conservation and efficient plant maintenance are important requirements in homes, gardens, greenhouses and agricultural environments. Traditional plant watering methods depend on manual observation and fixed schedules, which may result in overwatering, underwatering and unnecessary water consumption. These problems can be reduced by introducing an automated irrigation system that monitors the actual moisture condition of the soil.

This project presents the design and development of a low-cost Smart Plant Watering System using an Arduino Uno microcontroller, a soil moisture sensor, a relay module and a small DC water pump. The sensor measures the moisture level in the soil and sends the readings to the Arduino. The controller compares the sensor values with a predefined threshold and activates the pump when the soil becomes dry. When sufficient moisture is detected, the pump is switched off automatically.

The system is designed to reduce human intervention, improve watering consistency and minimize unnecessary water usage. It can be implemented using commonly available electronic components and can be extended with Internet of Things (IoT) technology for remote monitoring and control. The proposed prototype is suitable for potted plants, home gardens, nurseries and small-scale agricultural applications.

The system can be evaluated through sensor calibration, automatic switching tests, pump response measurements and repeated dry-soil and moist-soil trials. The design provides a practical demonstration of sensor-based automation and efficient water management.

Keywords: Smart plant watering, soil moisture sensor, Arduino Uno, automatic irrigation, water pump, relay module, water conservation, IoT.

3. Introduction

3.1 Background and Problem Definition

Water is essential for plant growth, nutrient absorption and agricultural productivity. Plants require an appropriate amount of water to maintain healthy growth. However, manual watering requires regular attention and may not always provide the correct quantity of water.

In homes, hostels, gardens and nurseries, plants may remain without water when people are busy or away. At the same time, watering plants without checking soil moisture can result in excess water consumption. Overwatering may cause poor soil aeration and root damage, whereas underwatering can cause plants to wilt and grow poorly.

Conventional plant watering methods have several limitations:

- Manual watering requires regular human involvement.
- Fixed watering schedules do not consider the actual moisture condition of the soil.
- Excessive watering can waste water and affect plant health.
- Plants may become dry when watering is delayed.
- Monitoring several plants individually can be time-consuming.
- Conventional systems may not provide automatic control or real-time monitoring.

Problem statement: To design and develop a low-cost Smart Plant Watering System that measures soil moisture and automatically operates a water pump to provide water when required, thereby reducing manual effort and unnecessary water consumption.

3.2 Literature Review

Several existing technologies and approaches provide the foundation for this project.

1. **Traditional irrigation systems:** Conventional irrigation methods include watering cans, hoses, sprinklers and drip irrigation. These methods are widely used but may require manual operation or scheduled control and may not respond directly to changes in soil moisture.
2. **Timer-based irrigation systems:** Timer-based systems operate pumps or valves at predefined intervals. They reduce manual effort but may supply water even when the soil is already sufficiently moist.
3. **Soil moisture sensor-based systems:** Soil moisture sensors measure the moisture condition of the soil. Their readings can be used by a controller to determine when watering is necessary. Sensor calibration is important because readings depend on soil type, sensor construction and installation conditions.
4. **Microcontroller-based irrigation:** Controllers such as Arduino Uno can process sensor readings and operate pumps through relay modules or motor drivers. These systems are affordable and suitable for educational projects, home gardens and small-scale irrigation.
5. **IoT-based smart irrigation:** Wi-Fi-enabled controllers such as ESP32 can transmit moisture readings to online dashboards and mobile applications. These systems support remote monitoring and can be extended with notifications and environmental data.

Research gap: Manual and timer-based systems do not always respond to the actual moisture condition of the soil. A simple, low-cost sensor-based watering system provides an accessible solution by combining soil moisture measurement, automatic pump control and the possibility of future IoT integration.

3.3 System Architecture

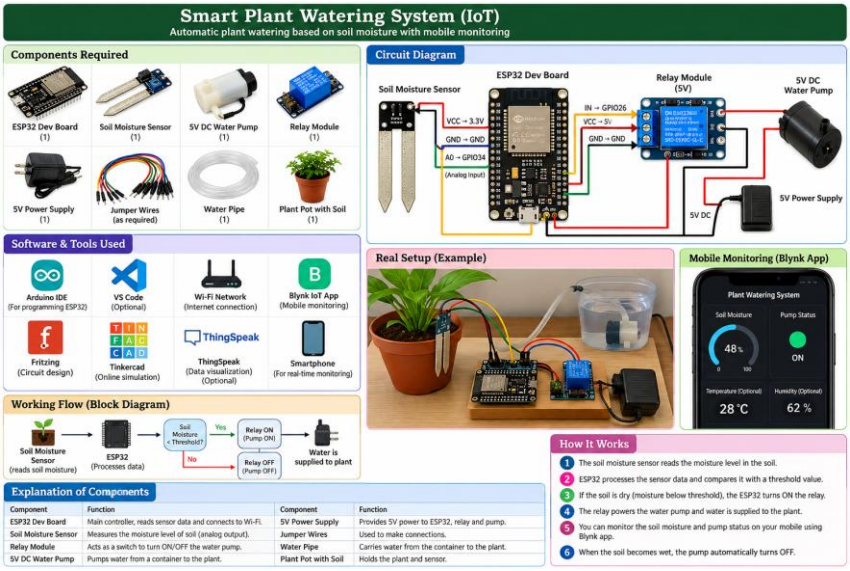


Figure 3.1: Overall Architecture and Working Flow of the Smart Plant Watering System

Working in brief:

Condition	Sensor reading	Controller action	Pump status
Soil is dry	Dry-soil range	Activates driver	ON
Soil is sufficiently moist	Moist-soil range	Deactivates driver	OFF
Sensor reading changes	Updated value	Re-evaluates threshold	Automatic control
Power supply is unavailable	No valid operation	Controller stops	OFF

The exact sensor readings and comparison direction must be established through calibration before operation.

4. Objective and Methodology

4.1 Objectives

1. To design a low-cost automatic plant watering system using a soil moisture sensor.
2. To measure the moisture condition of the soil continuously.
3. To activate a water pump automatically when the soil becomes dry.
4. To stop the pump when sufficient moisture is detected.
5. To reduce unnecessary water consumption and manual effort.
6. To develop a simple and easily maintainable irrigation prototype.
7. To evaluate the system through functional and performance testing.
8. To provide a foundation for future IoT-based plant monitoring.

4.2 Methodology

The project follows a systematic hardware development process consisting of requirement analysis, component selection, circuit design, programming, prototype assembly, calibration and testing.

Step 1 – Requirement analysis: The system must detect soil moisture, automatically operate a small water pump and stop watering when the soil reaches the selected moisture threshold. The design should be affordable, simple and suitable for small plants.

Step 2 – Component selection: An Arduino Uno is selected as the controller, a soil moisture sensor measures the soil condition, and a relay module or motor driver controls the pump. A suitable power supply, connecting wires, water container and tubing complete the setup.

Step 3 – Circuit design: The sensor output is connected to an analog input of the Arduino. The pump is connected through a driver circuit rated for its operating voltage and current. The controller and pump power arrangements are checked before testing.

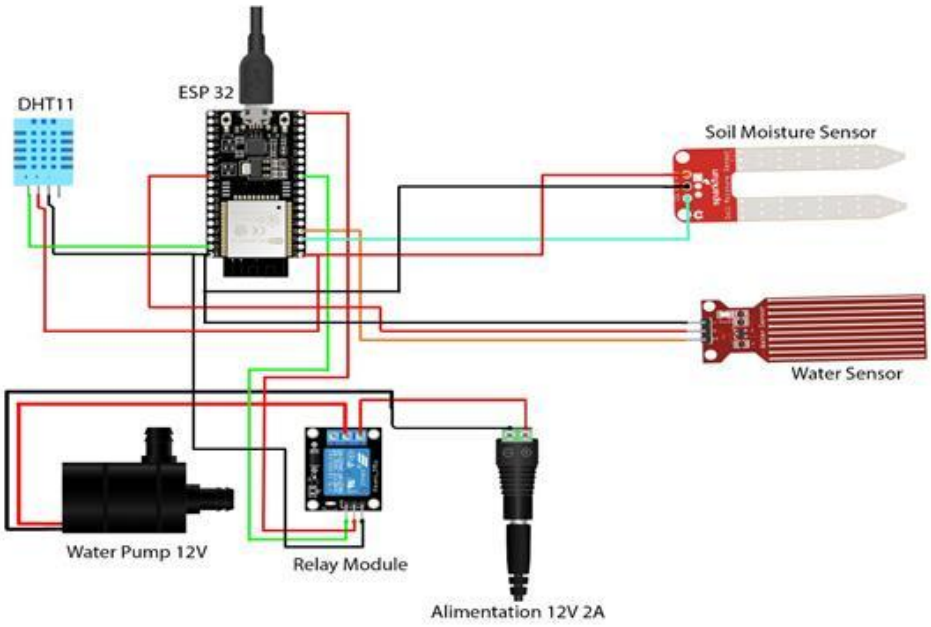


Figure 4.1: Circuit Connection Diagram of ESP32-Based Smart Plant Watering System

Step 4 – Programming: A program is developed to read the sensor value, compare it with the calibrated threshold and control the pump. The sensor reading and pump status can be displayed through the Arduino Serial Monitor.

Step 5 – Prototype assembly: The components are assembled on a breadboard or suitable mounting board. The moisture sensor is placed in the soil near the plant roots, and the pump tubing is positioned to deliver water to the plant.

Step 6 – Calibration and testing: Sensor values are recorded in dry and moist soil. A suitable threshold is selected, and the pump response is tested under different soil conditions.

Step 7 – Performance evaluation: The system is evaluated for automatic switching, sensor response, pump operation, repeatability, water delivery and reliability.

4.3 Hardware Components

S.No.	Component	Specification	Qty.	Purpose
1	Microcontroller	Arduino Uno	1	Processes sensor readings and controls the system
2	Soil moisture sensor	Analog output type	1	Measures soil moisture
3	Relay module / motor driver	Compatible with pump voltage and current	1	Controls the pump
4	DC water pump	Small low-voltage pump	1	Supplies water to the plant
5	Power supply	Suitable for controller and pump	1 set	Provides electrical power

6	Connecting wires	Jumper wires	1 set	Establishes circuit connections
7	Water container	Suitable size	1	Stores water
8	Water tubing	Compatible with pump outlet	1	Transfers water to the plant
9	Plant pot and soil	Suitable for the selected plant	1 set	Provides the growing environment
10	Breadboard / mounting board	General purpose	1	Supports circuit assembly

The approximate cost depends on the selected components and supplier. A basic prototype may cost around ₹500–₹1,200, excluding tools and any optional IoT hardware.

4.4 Software

The Arduino IDE is used to write, compile and upload the program. Embedded C/C++ is used to implement the sensor-reading and pump-control logic. The Serial Monitor is used to observe sensor readings and verify the operating status.

No internet connection is required for the basic Arduino-based implementation. IoT monitoring can be introduced later using a Wi-Fi-enabled controller and an appropriate dashboard.

4.5 Working Principle

Moisture sensing: The soil moisture sensor measures the moisture condition of the soil and provides an electrical output to the Arduino. Depending on the sensor design, the output may increase or decrease as the soil becomes drier.

Decision-making: The Arduino reads the sensor value and compares it with a predefined threshold. The threshold is selected after testing the sensor in dry and moist soil.

Automatic watering: When the sensor indicates that the soil is too dry, the Arduino activates the relay or motor driver. The pump draws water from the container and delivers it to the plant.

Automatic stopping: As the soil becomes sufficiently moist, the sensor reading changes. The controller switches off the pump to prevent unnecessary watering.

Continuous monitoring: The process repeats at regular intervals, enabling the system to respond to changes in soil moisture.

4.6 Sample Arduino Program

The following program assumes that a higher analog reading indicates drier soil. The threshold value is an example and must be calibrated for the sensor being used.

```
const int moisturePin = A0;

const int pumpPin = 7;

const int dryThreshold = 600;

void setup() {

  Serial.begin(9600);

  pinMode(pumpPin, OUTPUT);
```

```
 // Assumes an active-HIGH driver.
  digitalWrite(pumpPin, LOW);
}
void loop() {
  int moistureValue = analogRead(moisturePin);
  Serial.print("Soil Moisture: ");
  Serial.println(moistureValue);
if (moistureValue > dryThreshold) {
    digitalWrite(pumpPin, HIGH);
    Serial.println("Soil Dry - Pump ON");
  } else {
    digitalWrite(pumpPin, LOW);
    Serial.println("Soil Moist - Pump OFF");
  }
delay(1000);
}
```

The program initializes the sensor and pump-control pins. It then reads the moisture sensor, displays the reading and compares it with the selected threshold. The pump is activated when the soil is classified as dry and switched off otherwise.

The program must be adjusted if the sensor produces lower readings in dry soil or if the driver uses active-LOW control. A real implementation should also include appropriate pump run-time limits and protection against an empty water container.

5. Innovation, Ethical and Societal Impact

5.1 Innovation

- **Moisture-based automation:** Watering decisions are based on sensor measurements instead of relying entirely on fixed schedules.
- **Low-cost implementation:** Common electronic components make the system suitable for educational and small-scale applications.
- **Automatic pump control:** The controller operates the water pump without continuous human intervention.
- **Expandable architecture:** Additional sensors, water-level detection and wireless monitoring can be integrated into the design.
- **Efficient resource management:** The system aims to reduce unnecessary irrigation by supplying water when the soil requires it.

5.2 Safety Considerations

- The pump must be controlled through a suitable relay module or motor driver rather than directly from an Arduino output pin.
- The power supply must match the operating requirements of the controller and pump.
- Electrical connections should be protected from water and moisture.
- The pump should be prevented from running dry for extended periods.
- Tubing and electrical components should be arranged to avoid leakage near the circuit.
- The system should be tested with the appropriate electrical precautions before unattended operation.

5.3 Ethical Aspects

- **Responsible water usage:** The project encourages efficient water management and helps reduce avoidable wastage.
- **Honest reporting:** Sensor readings, water savings and performance measurements should be reported from actual observations rather than assumed values.
- **Responsible electronic waste management:** Damaged sensors, wires and electronic components should be disposed of through appropriate collection facilities.
- **Transparency:** Users should understand the limitations of the sensor and the need for periodic calibration and maintenance.

5.4 Societal and Environmental Impact

- **Water conservation:** Moisture-based control can reduce unnecessary watering when the system is correctly calibrated.
- **Support for plant care:** The system can help maintain plants in homes, hostels, offices and educational institutions.
- **Reduced manual effort:** Automatic operation is useful for users who manage several plants or cannot water them regularly.
- **Environmental awareness:** The project demonstrates how automation can contribute to more responsible resource use.
- **Educational value:** Students can learn about sensors, microcontrollers, embedded programming and automated control through practical implementation.

5.5 Real-world Applicability

The Smart Plant Watering System can be used in home gardens, balcony gardens, indoor plant arrangements, nurseries, greenhouses and small agricultural plots. With suitable modifications, the system can support multiple plants, drip irrigation, remote monitoring and automated watering schedules.

6. Result and Discussion

6.1 Test Setup

The proposed prototype consists of an Arduino Uno, a soil moisture sensor, a compatible pump driver, a DC water pump, a power supply, a water container and a plant pot containing soil.

The sensor is placed in the soil near the plant roots. The Arduino reads the sensor output and compares it with the calibrated threshold. The pump is tested under dry-soil and moist-soil conditions to verify the automatic control process.

Note to students: The values and observations below are intended as a testing template. Record actual readings from your own prototype before submission. Do not report unperformed tests as completed experiments.

6.2 Functional Test

Test	Expected behaviour	Observed behaviour	Result
Sensor placed in dry soil	Pump activates	Record actual observation	Pending testing
Sensor placed in moist soil	Pump stops	Record actual observation	Pending testing
Soil moisture changes	Controller updates pump status	Record actual observation	Pending testing
Sensor readings displayed	Values appear in Serial Monitor	Record actual observation	Pending testing
Water container becomes empty	Pump protection responds if fitted	Record actual observation	Pending testing

6.3 Performance Analysis

The following parameters can be recorded during evaluation.

Parameter	Measurement method
Dry-soil sensor reading	Record analog reading in dry soil
Moist-soil sensor reading	Record analog reading in moist soil
Switching response	Measure the time between threshold detection and pump activation
Pump operating time	Record the duration of each watering cycle
Water delivered	Measure the volume supplied during a test
Repeatability	Repeat the dry and moist soil tests
Power consumption	Measure voltage and current using suitable instruments

The difference between dry-soil and moist-soil sensor readings helps determine an appropriate threshold. The pump should switch on when the soil reaches the dry condition and switch off when the required moisture level is achieved.

6.4 Discussion

The proposed system provides a simple method of automatic irrigation by connecting soil moisture measurement with pump control. Unlike a fixed timer, it uses the measured soil condition to make the watering decision.

The effectiveness of the system depends on sensor calibration, soil type, sensor placement, pump capacity and the amount of water delivered. Repeated tests are required to establish a reliable threshold. Adding a water-level sensor and a maximum pump run-time limit would improve the safety and reliability of the design.

Water savings should be determined by comparing the quantity of water used by the automated system with a suitable manual or timer-based baseline under comparable conditions.

6.5 Limitations

- A single sensor may not represent the moisture condition of a large planting area.
- Sensor readings vary with soil composition and installation conditions.
- The basic system does not automatically identify plant species or their specific water requirements.
- An empty water container may prevent successful watering.
- Sensor failure or incorrect calibration can cause inappropriate pump operation.
- The basic Arduino version does not provide remote monitoring without additional hardware.

7. Conclusion

The Smart Plant Watering System presents a practical and economical approach to automating plant irrigation. The design uses a soil moisture sensor to monitor soil conditions and an Arduino controller to operate a water pump when watering is required. This approach reduces dependence on manual watering and can help avoid unnecessary water consumption.

The project demonstrates the application of sensors, embedded programming and electronic switching to a real-world problem. Its modular design makes it suitable for educational demonstrations and small-scale plant care applications. Actual performance depends on sensor calibration, correct electrical connections and appropriate pump selection.

The project also provides a foundation for developing more advanced irrigation solutions that incorporate wireless connectivity, environmental monitoring and intelligent control techniques.

Future Scope

1. **IoT integration:** Use an ESP32 or similar controller to display soil moisture readings on a web dashboard or mobile application.
2. **Mobile notifications:** Send alerts when the soil is too dry, the water tank is empty or the pump encounters a fault.
3. **Multiple sensor support:** Monitor different pots or irrigation zones independently.
4. **Water-level detection:** Add a tank-level sensor to prevent pump operation when water is unavailable.
5. **Solar-powered operation:** Introduce a suitable solar panel and battery system for outdoor applications.
6. **Advanced irrigation control:** Use moisture ranges, pump run-time limits and calibrated watering cycles to improve control.
7. **Weather integration:** Incorporate rainfall information and weather forecasts to avoid unnecessary irrigation.

8. **Smart agriculture extension:** Combine soil moisture with temperature, humidity and other environmental measurements for improved plant management.

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5. Food and Agriculture Organization of the United Nations, *Water for Sustainable Food and Agriculture*. FAO, Rome.
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7. Refer to the datasheet supplied by the manufacturer of the soil moisture sensor selected for the prototype.
8. Refer to the datasheet supplied by the manufacturer of the selected relay module or motor driver and DC water pump.

Note: Verify the component specifications and bibliographic details of any additional references used in the final submission.